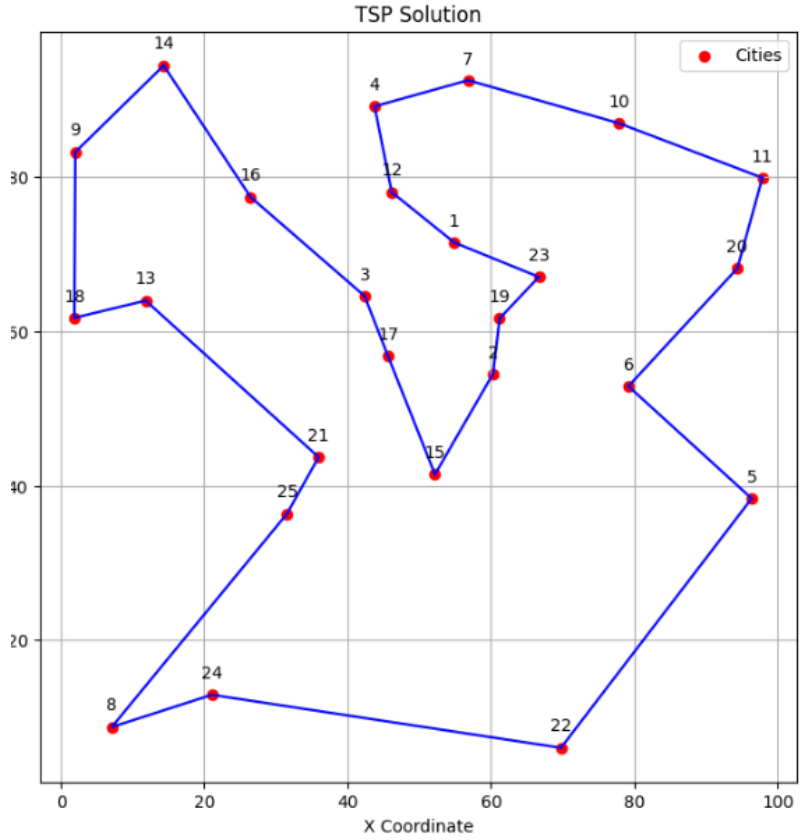
Vehicle Routing Problem (VRP)

Travelling Salesman Reminder

- Finding the shortest possible tour that visits each city exactly once and returns to the original city.

- A set of Vertices $i \in V$
- A set of Edges $(i, j) \in E$
- $x_{ij} \in \{0,1\}$ - Edge (i, j) chosen or not
- c_{ij} - Cost of edge (i, j)



MILP

$$\min. \sum_{(i,j)\in E} c_{ij}x_{ij}$$

$$s.t.$$

$$\sum_{j=1:i\neq j}^{n} x_{ij} = 1 \quad \forall i \in \{1, \dots n\}$$

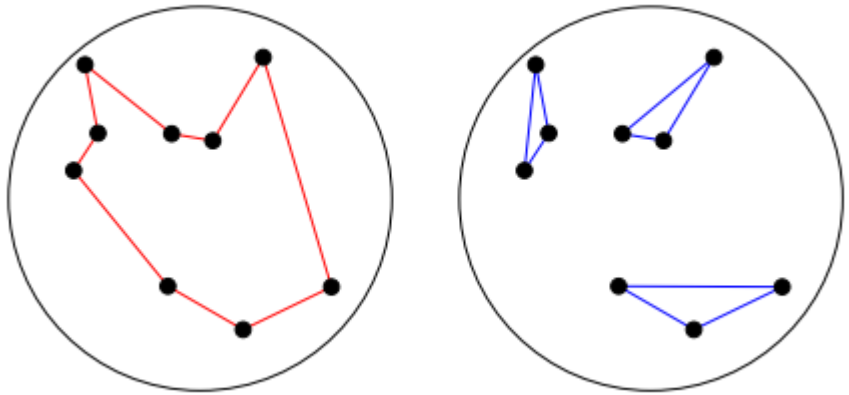
$$\sum_{j=1:i\neq j}^{n} x_{ji} = 1 \quad \forall i \in \{1, \dots n\}$$

$$x_{ij} \in \{0,1\}$$

Whilst all cities will be visited, it doesn't guarantee a single tour!

We could have lots of subtours that cover all cities.

Subtour example



$$\min. \sum_{(i,j)\in E} c_{ij}x_{ij}$$

$$s.t.$$

$$\sum_{j=1:i\neq j}^{n} x_{ij} = 1 \quad \forall i \in \{1, \dots n\}$$

$$\sum_{j=1:i\neq j}^{n} x_{ji} = 1 \quad \forall i \in \{1, \dots n\}$$

$$x_{ij} \in \{0,1\}$$

Both the red (complete tour) and the blue (subtours) have exactly one edge entering and one edge leaving a city!

TSP MTZ Constraint

- Let $u_i \in \mathbb{R}$ represent the order in which city i is visited

- u_i need not equal the position: only the order of the values matters.
- What we are looking for is some increasing sequence $u_{r_1} < u_{r_2} < \cdots < u_{r_n}$, where r_k is the city visited k-th, i.e. $r_2 = 3$ means that city 3 is visited second.

- If city i is visited just before city j then we would expect the following:

- $x_{ij} = 1$ (because we go from city i to city j)
- $u_j > u_i$ (because we visit city i before city j)

TSP MTZ Constraint

If city i is visited just before city j then we would expect the following:

- $x_{ij} = 1$ (because we go from city i to city j)
- $u_j > u_i$ (because we visit city i before city j)

Hence we require that:

If $x_{ij} = 1$

then $u_j > u_i$

TSP MTZ Constraint

$$u_i - u_j + 1 \leq M(1 - x_{ij})$$

1 is a common choice, it could be any positive number…

Case $x_{ij} = 0$:

$$u_i - u_j + 1 \leq M$$

Automatically satisfied as long as the difference between $u_i - u_j \leq M - 1$

Case $x_{ij} = 1$

$$u_i - u_j + 1 \leq 0 \implies u_j \geq u_i + 1 \implies u_j > u_i$$

As required!

Why does this work?

We can continue this reasoning.

We know that there must exist an r_2 such that $x_{1r_2} = 1$ (i.e. we leave city 1).

There also must be $x_{r_2r_3} = 1$, and $x_{r_3r_4} = 1$, and $x_{r_4r_5} = 1$, and …

Thus, $u_{r_2} < u_{r_3} < u_{r_4} < u_{r_5} < \cdots < u_{r_n}$.

Hence there will be an increasing sequence of u_i, thus precluding $u_i = u_j$.

Why does this work?

Once this sequence is in place.

$$u_{r_2} < u_{r_3} < u_{r_4} < u_{r_5} < \cdots < u_{r_n}$$

Then we have to choose some k such that:
$x_{k1} = 1$ i.e. some edge enters city 1.

However, the only remaining city which doesn't have an edge leaving is r_n.

Thus, $k = r_n$, i.e., $x_{r_n 1} = 1$

This is valid if we do not include city 1 in the MTZ constraint. Hence, we have a tour $1, r_2, r_3, r_4, \ldots, r_n, 1$

TSP MTZ Constraint

$$u_i - u_j + 1 \leq M(1 - x_{ij})$$

How do we pick M sufficiently large?

<u>Simple way (non-optimal)</u>

Pick a very big number

TSP MTZ Constraint

$$u_i - u_j + 1 \leq M(1 - x_{ij})$$

How do we pick M sufficiently large?

<u>Optimal</u>

Along the tour (after city 1) there will be a sequence $u_{r_2} < u_{r_3} < u_{r_4} < u_{r_5} < \cdots < u_{r_n}$.

Thus simultaneously:

$$\begin{aligned} u_{r_2} - u_{r_3} + 1 &\leq 0 \\ u_{r_3} - u_{r_4} + 1 &\leq 0 \\ &\vdots \\ u_{r_{n-1}} - u_{r_n} + 1 &\leq 0 \end{aligned}$$

Add these together and we have $u_{r_2} - u_{r_n} + (n-2) \leq 0$.

Thus, the biggest difference $u_{r_n} - u_{r_2} \geq n - 2 \Longrightarrow u_{r_n} - u_{r_2} + 1 \geq n - 1$

We also know there is no edge between r_n and r_2 thus

$$\begin{aligned} u_{r_2} - u_{r_n} + 1 &\leq M \\ u_{r_n} - u_{r_2} + 1 &\leq M \end{aligned}$$

The first is satisfied if $M \geq 1$

The second is satisfied if:

$$M \geq u_{r_n} - u_{r_2} + 1 \geq n - 1$$

Hence pick $M \geq n - 1$.

Compact with Set Notation

Minimise:

$$\sum_{(i,j)\in E} c_{ij}x_{ij}$$

Subject to:

$$\sum_{i\in V:i\neq j} x_{ij} = 1 \qquad \forall j \in V$$

$$\sum_{j\in V:j\neq i} x_{ij} = 1 \qquad \forall i \in V$$

$$u_i - u_j + 1 \leq (n-1)\big(1 - x_{ij}\big) \qquad \forall i,j \in V \setminus \{1\}: i \neq j$$

$$x_{ij} \in \{0,1\}$$

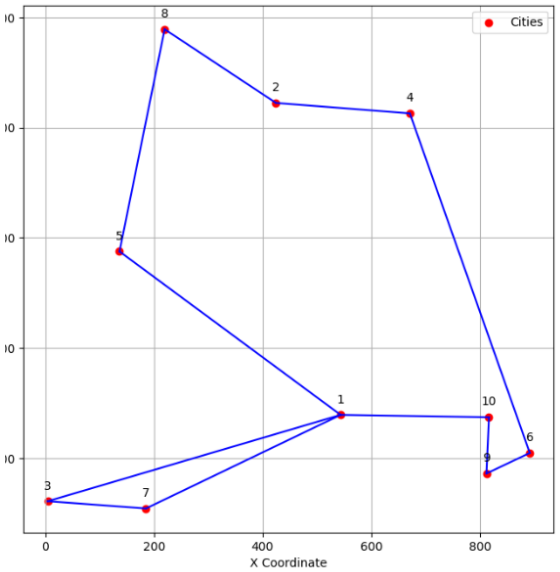
Extending the TSP to Allow Multiple Tours

Let's say we want m sub-tours.

Relax incoming and outgoing edge constraints for city 1

$$\sum_{i \in V : i \neq j} x_{ij} = 1 \qquad \forall j \in V \setminus \{1\}$$

$$\sum_{j \in V : j \neq i} x_{ij} = 1 \qquad \forall i \in V \setminus \{1\}$$



Add in that the number of edges leaving city 1 is equal to m

$$\sum_{j \in V : j \neq 1} x_{1j} = m$$

Extending the TSP to Allow Multiple Tours

Minimise:

$$\sum_{(i,j)\in E} c_{ij}x_{ij}$$

Subject to:

$$\sum_{i\in V:i\neq j} x_{ij} = 1 \qquad \forall j \in V \setminus \{1\}$$

$$\sum_{j\in V:j\neq i} x_{ij} = 1 \qquad \forall i \in V \setminus \{1\}$$

$$\sum_{j\in V:j\neq 1} x_{1j} = m$$

$$u_i - u_j + 1 \leq (n-1)\left(1 - x_{ij}\right) \qquad \forall i,j \in V \setminus \{1\}: i \neq j$$

$$x_{ij} \in \{0,1\}$$

$V \setminus \{1\}$ is the set V without 1, e.g. for n cities, $V = \{1,2,\dots,n\}$ and $V \setminus \{1\} = \{2,\dots,n\}$

Why does this work?

We have removed the need to pick a single edge leaving location 1 and entering location 1.

Therefore, we no longer need one single chain through all the cities, i.e. $u_{r_2} < u_{r_3} < u_{r_4} < u_{r_5} < \cdots < u_{r_n}$

Instead, we can build several chains and join both ends of each chain to location 1.

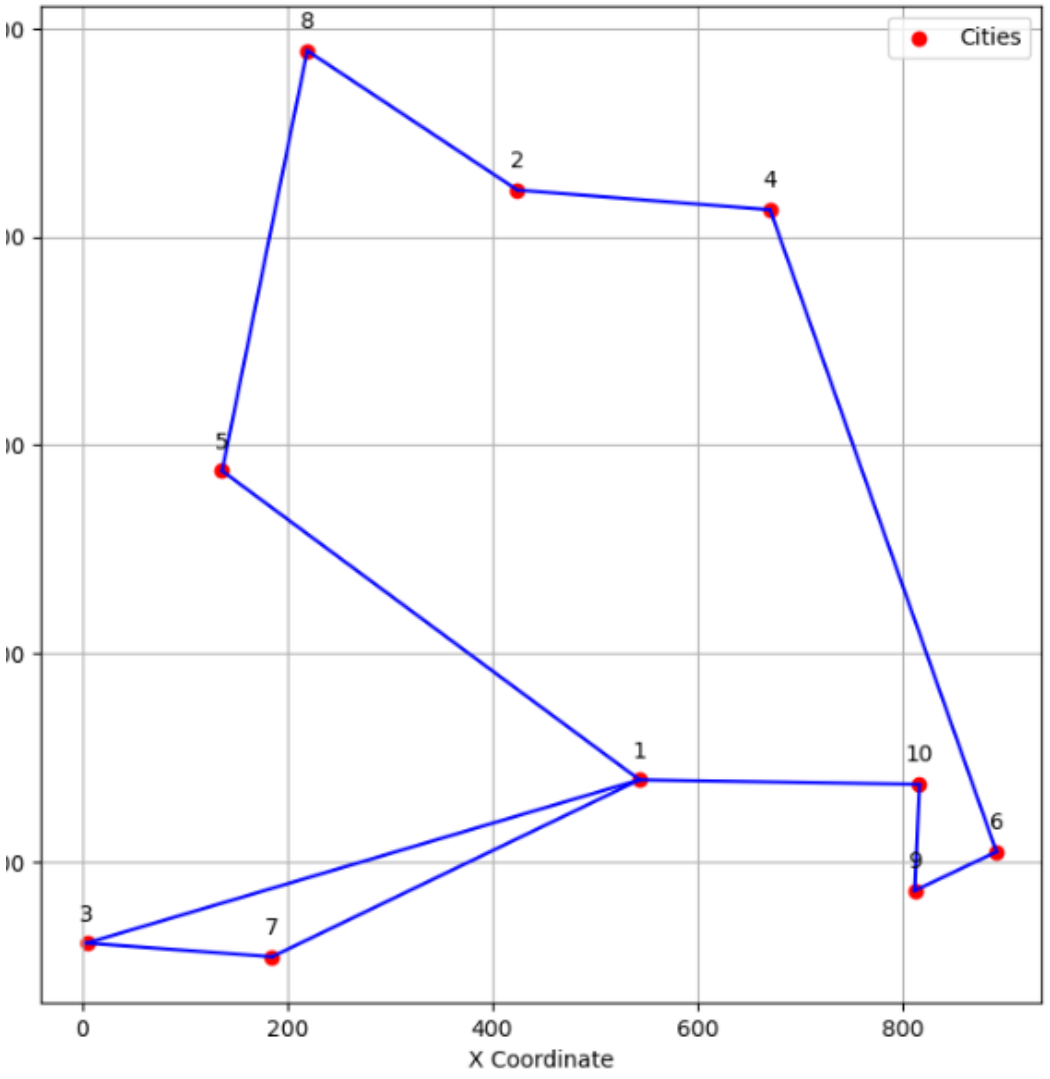
Note we could have any number of chains, including a single one. E.g. TSP solution.

Illustration

We might have two chains, each joined to location 1 at both ends (routes 1 → 7 → 3 → 1 and 1 → 5 → … → 10 → 1):

$$u_7 < u_3$$

$$u_5 < u_8 < u_2 < u_4 < u_6 < u_9 < u_{10}$$



Basic VRP Definition (Woolly)

What is the optimal set of routes for a fleet of m vehicles to deliver to a given set of V customers?

If $m = 1$ then we have the Travelling Salesman Problem!

What isn't defined in the above definition?

What is the objective?

What is the **optimal** set of routes for a fleet of m vehicles to deliver to a given set of V customers?

What do we mean by optimal?

- Minimise total distance travelled
- Minimise distance of longest route (i.e. minimise time taken for all deliveries)

Demand and capacity?

Demand

There is no demand for the delivery? This might be OK for the Royal Mail, but you can imagine a certain amount that needs to be shipped to a location.

Even for the Royal Mail, if every delivery is one uniform parcel, the demand is roughly one parcel per location.

Capacity

There is no capacity for a vehicle (route).

No idea if the route can carry the demand needed. i.e. is the van big enough?

Minimise the total distance

Objective

Minimise the sum of distance for the edges chosen.

Parameters

c_{ij} - distance (cost) between i and j

V – Set of cities (nodes)

Constraints

Each route is a subtour starting at the base depot. We will call this location 1.

There will be $m \geq 1$ subtours (i.e. m vehicles used).

We don’t care about demand or capacity

Minimise the total distance

Relax the depot constraint: at most m edges leave city 1

Minimise:

$$\sum_{(i,j)\in E} c_{ij}x_{ij}$$

Subject to:

$$\sum_{i\in V:i\neq j} x_{ij} = 1 \qquad \forall j \in V \setminus \{1\}$$

$$\sum_{j\in V:j\neq i} x_{ij} = 1 \qquad \forall i \in V \setminus \{1\}$$

$$\sum_{j\in V:j\neq 1} x_{1j} \leq m$$

$$u_i - u_j + 1 \leq (n-1)\big(1 - x_{ij}\big) \qquad \forall i,j \in V \setminus \{1\}: i \neq j$$

$$x_{ij} \in \{0,1\}$$

$V \setminus \{1\}$ is the set V without 1, e.g. for n cities, $V = \{1,2,\dots,n\}$ and $V \setminus \{1\} = \{2,\dots,n\}$

Pros and Cons

It will find you the optimal number of vehicles to minimise distance travelled.

It does not have capacity or demand built in.

Therefore, if the distances obey the triangle inequality, a single vehicle (the TSP tour) is always one of the best solutions.

Minimise total distance with capacity and demand

Issue!

- We can get long routes for some vans, short for others.
- This might be fine, but it is likely that a company would like to deliver things quickly.

Full Notation

- V – Set of locations (city 1 is the depot)
- m – number of vehicles
- q_i - demand of customer $i \in V$
- Q is the capacity of each vehicle
- c_{ij} - distance between customer i and customer j
- $x_{ij} \in \{0,1\}$ - decision variable indicating whether a vehicle travels from city i to city j.
- f_{ij} - Flow carried from city i to city j
  - What is in the van at that point in time!

Minimise total distance with capacity and demand

Objective:

$$\text{Minimise} \quad \sum_{i \in V} \sum_{j \in V : j \neq i} c_{ij} x_{ij}$$

Subject to:

1. Each city is left exactly once (excluding city 1):

$$\sum_{j \in V : j \neq i} x_{ij} = 1, \quad \forall i \in V \setminus \{1\}$$

2. Each city is visited exactly once (excluding city 1):

$$\sum_{i \in V : i \neq j} x_{ij} = 1, \quad \forall j \in V \setminus \{1\}$$

3. Vehicles leaving depot

$$\sum_{j \in V} x_{1j} \leq m$$

4. Flow conservation maintained

$$\sum_{j \in V} f_{ji} - \sum_{j \in V} f_{ij} = q_i, \quad \forall i \in V \setminus \{1\}$$

5. The load on an arc is at most the vehicle capacity Q

$$0 \leq f_{ij} \leq Q x_{ij}$$

$$x_{ij} \in \{0, 1\}$$

$$f_{ij} \in \mathbb{R}$$

Why didn't we include the sub-tour constraints?

To see why consider a sub-tour that does not include city 1. i.e. $r_i, \dots r_p, r_i$

$$
\begin{aligned}
f_{r_i r_j} - f_{r_j r_k} &= q_{r_j} && \text{Flow conservation for } r_j \\
f_{r_j r_k} - f_{r_k r_l} &= q_{r_k} && \text{Flow conservation for } r_k \\
&\vdots \\
f_{r_p r_i} - f_{r_i r_j} &= q_{r_i} && \text{Flow conservation for } r_i
\end{aligned}
$$

Adding all of these together results in:

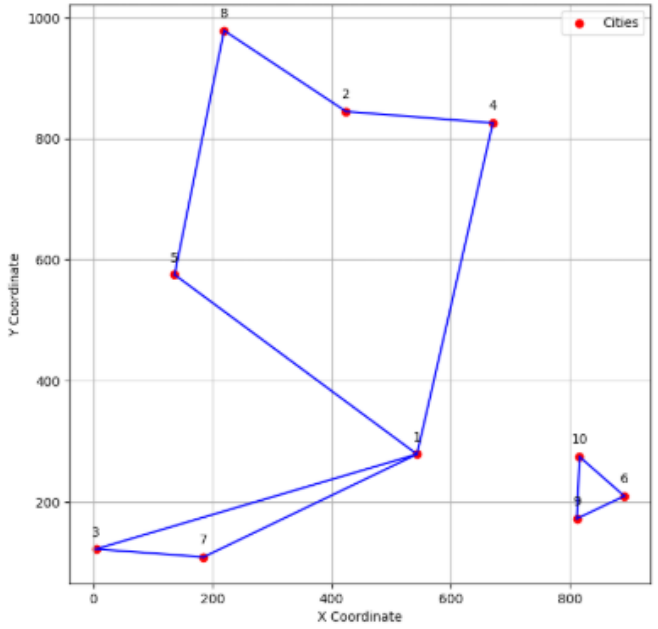
$$0 = q_{r_i} + q_{r_j} + \cdots + q_{r_p}$$

Not possible if every customer has $q_i > 0$. A subtour needs some demand ≤ 0: a pick-up or a stop with no demand.

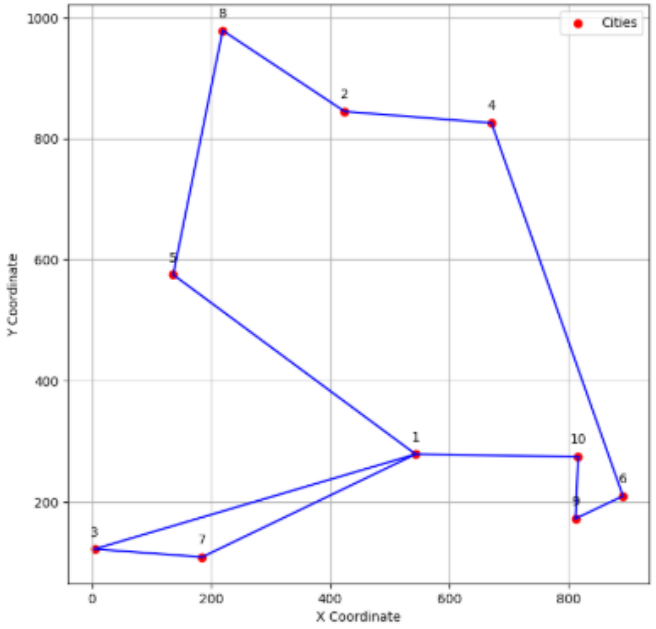
Cooked up example

$$u_i - u_j + 1 \leq (n-1)\left(1 - x_{ij}\right) \qquad \forall i, j \in V \setminus \{1\}: i \neq j$$

A cooked-up example where it is needed: a flow model in which some customers' demands add up to zero (pick-ups or zero demands).



Without sub tour constraints

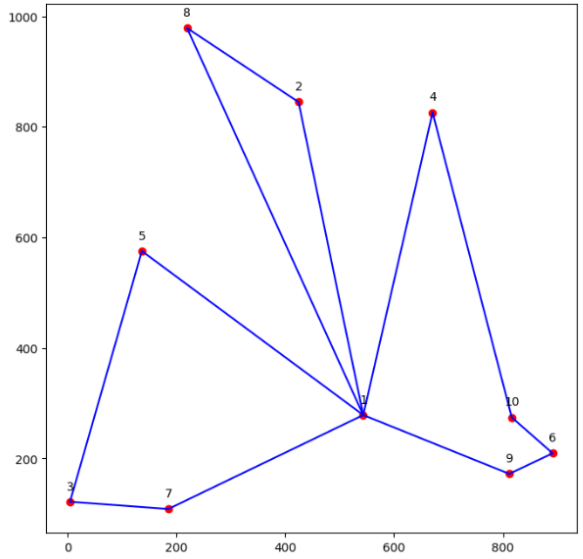


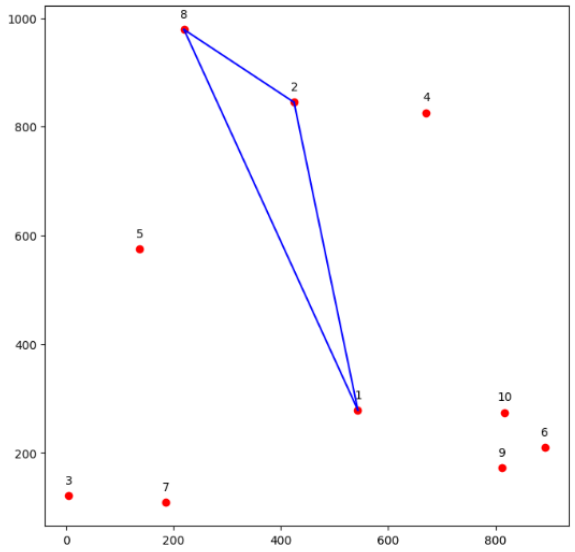
With sub tour constraints

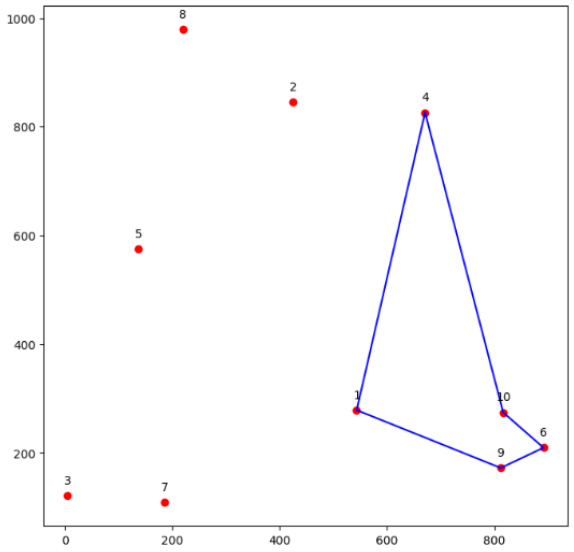
Deliver all orders as quickly as possible

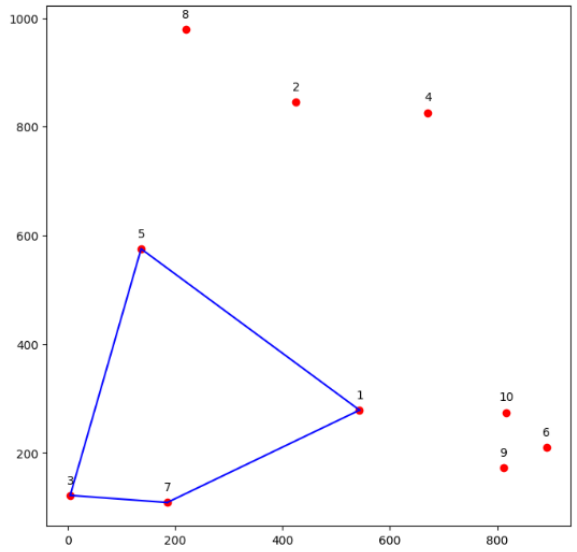
- We can reframe this objective as minimise the maximum route length of a vehicle.
- This is known as a minimax problem.
- To do this we will need to know the length of each of the routes for the vehicles
- A nice way to visualise this idea is to imagine we are creating a copy of the network for each vehicle
  - We can then track which edges are used in each of the networks.
  - We can work out the length of each tour
  - Minimise the maximum length of all tours considered.

Illustration









Minimise the distance of the longest route

Objective

Minimise the distance of the longest route

- This is the same as trying to finish all of the orders ASAP. Is this better?

Parameters

c_{ij} - distance between i and j

V – Set of cities (nodes)

Each location i has a demand q_i

Each vehicle has a capacity Q

Constraints

Each route is a subtour starting at the base depot. We will call this location 1.

There are at most m subtours (i.e. at most m vehicles used).

Demand q_i delivered to i

Route demand does not exceed capacity.

Full Notation

- V – Set of locations (city 1 is the depot)
- K – Set of vehicles
- m – number of vehicles that may leave the depot
- q_i - demand of customer $i \in V$
- Q is the capacity of each vehicle
- c_{ij} - distance between customer i and customer j
- $x_{ijk} \in \{0,1\}$ - decision variable indicating whether vehicle k travels from city i to city j.
- f_{ijk} - Flow carried from city i to city j by vehicle k
  - What is in the van at that point in time!

Full Model

Minimise: C_{Max}

Subject to:

$$\sum_{k \in K} \sum_{j \in V : j \neq i} x_{ijk} = 1, \quad \forall i \in V \setminus \{1\}$$ A single vehicle leaves a location

$$\sum_{k \in K} \sum_{i \in V : i \neq j} x_{ijk} = 1, \quad \forall j \in V \setminus \{1\}$$ A single vehicle enters a location

$$\sum_{j \in V : j \neq i} x_{jik} = \sum_{j \in V : j \neq i} x_{ijk}, \quad \forall i \in V \setminus \{1\}, \forall k \in K$$ If vehicle k enters location i then it must leave location i

$$\sum_{k \in K} \sum_{j \in V} x_{1jk} \leq m$$ No more than m vehicles leave the depot

$$\sum_{i \in V} \sum_{j \in V : i \neq j} c_{ij} x_{ijk} \leq C_{Max}, \quad \forall k \in K$$ Distance of route for a vehicle is at most C_{Max}

$$\sum_{k \in K} \sum_{j \in V} f_{jik} - \sum_{k \in K} \sum_{j \in V} f_{ijk} = q_i, \quad \forall i \in V \setminus \{1\}$$ Enough demand is delivered to a location

$$0 \leq f_{ijk} \leq Q x_{ijk}, \quad \forall i, j \in V, \forall k \in K : i \neq j$$ The load on an arc is at most the vehicle capacity

$$x_{ijk} \in \{0, 1\}, \quad \forall i, j \in V, \forall k \in K : i \neq j$$

$$f_{ijk} \in \mathbb{R}, \quad \forall i, j \in V, \forall k \in K : i \neq j$$

Optional: sub-tour constraints. Include them if solving without them gives sub-tours that do not start at city 1.

$$u_{ik} - u_{jk} + 1 \leq (n-1)(1 - x_{ijk}), \quad \forall i, j \in V \setminus \{1\}, \forall k \in K : i \neq j$$ MTZ sub-tour constraint for each vehicle k

Vehicle Routing on a Subgraph

Route m vehicles to visit a set of locations S subset of V.

Do we need the whole graph?

No

We could compute the shortest path between the locations, this would then give us a distance matrix and we can solve as standard using just the locations S.

You can imagine a logistics company doing this each day, working out the distances between delivery locations and then running a VRP model.

Conclusion

Discrete Optimisation is Cool!

It is also a fact of life!

Interactive Notebook

The demonstration notebook runs in your browser on the course site: Week 10, 2 Lecture Demos, 1 TSP and VRP with Pyomo.ipynb.